Efficacy and safety of once-daily oral zenagamtide, a novel unimolecular GLP-1 and amylin receptor agonist, in adults with type 2 diabetes: a multicentre, randomised, parallel, double-blind, placebo-controlled, dose-finding, phase 2 trial

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Summary

Background Zenagamtide (formerly amycretin) is a unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors. We aimed to investigate the dose–response relationship with respect to the efficacy and safety of once-daily oral zenagamtide compared with placebo in adults with type 2 diabetes.

Methods This 36-week, randomised, parallel, double-blind, placebo-controlled, dose-finding, phase 2 trial was conducted at 83 hospital and clinic sites across 11 countries. The trial consisted of a 3-week screening period, a 36-week intervention period, and a 4-week follow-up period. Adults (aged 18–75 years) with type 2 diabetes (glycated haemoglobin [HbA_{1c}] 7·0–10·0% [53–86 mmol/mol]), treated with stable doses of metformin with or without a SGLT2 inhibitor, and with a BMI of 23·0 to <50·0 kg/m² were randomly assigned (5:1) to oral zenagamtide or placebo using a randomisation and trial supplies management system, then further randomly assigned (1:1:1) to one of three dose groups (6, 25, or 50 mg). Participants assigned to placebo were similarly assigned to matched placebo regimens corresponding to each dosing schedule. Randomisation was stratified by HbA_{1c} (<8·5% or ≥8·5%) at screening and country of residence (Japan or other countries). The starting dose of oral zenagamtide was 1·5 mg, with dose escalations every 4 weeks until the maintenance dose was reached (6, 25, or 50 mg). Participants and staff were masked within each dose level to placebo or zenagamtide. The primary outcome was change in HbA_{1c} from baseline to week 36. The primary outcome was assessed in all randomly assigned participants using the efficacy estimand based on on-treatment data without rescue medication. Safety outcomes were assessed in all randomly assigned participants exposed to treatment. This trial is registered with ClinicalTrials.gov, NCT06542874, and is completed.

Findings Between Aug 7 and Dec 27, 2024, 186 (20%) of 915 screened participants were randomly assigned to oral zenagamtide (54 participants to 6 mg, 51 participants to 25 mg, and 51 participants to 50 mg) or placebo (30 participants) and were included in the full analysis set. At week 36, from baseline (mean range 7·9–8·1%), estimated mean change in HbA_{1c} was –0·9% with zenagamtide 6 mg (estimated treatment difference [ETD] *vs* placebo –0·5% [95% CI –1·04 to –0·04]; p=0·033), –1·3% with zenagamtide 25 mg (–0·99% [–1·49 to –0·49]; p=0·0001), and –1·4% with zenagamtide 50 mg (–1·09% [–1·59 to –0·59]; p<0·0001). The most common adverse events were gastrointestinal, occurring in 14 (26%) of 54 participants in the zenagamtide 6 mg group, 21 (41%) of 51 participants in the zenagamtide 25 mg group, 24 (47%) of 51 participants in the zenagamtide 50 mg group, and seven (23%) of 30 participants in the placebo group. Serious adverse events were reported in seven (4%) of 186 participants receiving zenagamtide: two in the 6 mg group, two in the 25 mg group, and three in 50 mg group. No serious adverse events were reported in the placebo group. No deaths occurred during the trial.

Interpretation In people with type 2 diabetes, once-daily oral zenagamtide showed clinically meaningful and significant improvements in HbA_{1c} from baseline to week 36 compared with placebo at all three dose levels (6, 25, and 50 mg). The safety and tolerability of oral zenagamtide was consistent with other GLP-1 and amylin receptor agonists.

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Introduction

The majority of adults with type 2 diabetes are living with overweight or obesity.[1] Because weight loss is known to improve glycaemic outcomes in type 2 diabetes,[2] it is recommended as a target as part of a holistic approach to type 2 diabetes management.[2,3] Another consideration in managing type 2 diabetes is an individual's preference regarding the treatment administration route (oral *vs* injectable) because this could have an effect on a person's willingness to initiate treatment and to adhere

Research in context

Evidence before this study

GLP-1 and amylin receptor agonists have complementary mechanisms of action for providing glycaemic control, weight management, and improvements in cardiometabolic parameters. Combination therapy with GLP-1 and amylin receptor agonists could therefore provide greater glycaemic control and weight loss than the individual components alone in individuals with overweight or obesity, with or without type 2 diabetes. Literature searches were conducted using PubMed for studies in any language published between Feb 12, 2016, and Feb 13, 2026, using search terms including "glucagon-like peptide AND amylin AND type 2 diabetes" and "glucagon-like peptide-1 AND amylin AND obesity". Relevant publications identified included phase 3 clinical trials for the GLP-1 analogue, semaglutide, the amylin analogue, cagrilintide, and a cagrilintide and semaglutide combination therapy, CagriSema. The REDEFINE 2 trial of once-weekly subcutaneous cagrilintide and semaglutide combination therapy (cagrilintide 2·4 mg and semaglutide 2·4 mg) in individuals with overweight or obesity and type 2 diabetes resulted in superior reductions in bodyweight compared with placebo. The combination therapy also provided clinically relevant improvements in glycaemic parameters, including glycated haemoglobin (HbA_{1c}) and continuous glucose monitoring metrics, and improved cardiovascular risk markers.

Zenagamtide (formerly known as amycretin), a unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors, has been investigated as both subcutaneous and oral formulations in individuals with overweight or obesity without type 2 diabetes. In both the phase 1b/2a trial of the subcutaneous formulation and the phase 1 trial of the oral formulation, zenagamtide resulted in significantly greater weight loss than placebo. The oral formulation of zenagamtide also resulted in greater reductions in HbA_{1c} than placebo. These trials supported further investigation of both the subcutaneous and oral formulations of zenagamtide in a dose-finding phase 2 trial in individuals with type 2 diabetes.

Added value of this study

To our knowledge, this was the first dose-finding trial to investigate the efficacy and safety of once-daily oral zenagamtide (6, 25, and 50 mg) compared with placebo in adults with type 2 diabetes. All doses of zenagamtide showed clinically meaningful and significantly greater reductions in HbA_{1c}, supported by improvements in time in range (TIR; 3·9–10·0 mmol/L [70–180 mg/dL]) from baseline to week 36. There was an estimated mean change from baseline in HbA_{1c} of up to –1·4% with zenagamtide 50 mg at week 36. The observed mean percentage of TIR at week 36 ranged from 73·8% to 82·9% (equivalent to approximately 17 h 43 min to 20 h per day) for the zenagamtide doses. Significantly greater percentage changes in bodyweight were also observed with zenagamtide 25 mg and 50 mg, with no signs of plateauing. The largest estimated percentage change in bodyweight with zenagamtide from baseline to week 36 was –10·3% for the 25 mg dose. Zenagamtide had a safety and tolerability profile consistent with GLP-1 and amylin receptor agonists and the previous trial with oral zenagamtide. The most common adverse events were gastrointestinal and were mild to moderate in severity.

Implications of all the available evidence

The results from this trial provide evidence of the therapeutic potential of the unimolecular agonist, zenagamtide, and support further investigation of once-daily oral zenagamtide in phase 3 trials. Given that people have individual preferences on the mode of delivery of medications (oral *vs* subcutaneous), the continued development of the oral formulation of zenagamtide could help to support individual choice, potentially increasing treatment acceptance and adherence while improving glycaemic control, facilitating attainment of weight loss goals, and improving other cardiometabolic parameters.

to treatment.[2] Although highly efficacious therapies are available for subcutaneous administration, which supports absorption and reduces proteolytic degradation in the gastrointestinal tract, some individuals remain reluctant to initiate injectable treatment. This reluctance could postpone timely treatment initiation and subsequent intensification. For example, a study comparing oral formulations of semaglutide with subcutaneous formulations of semaglutide in type 2 diabetes found that treatment adherence to the subcutaneous formulation was lower than adherence to the oral formulation,[4] highlighting the relevance of developing both oral and subcutaneous formulations of new glucose-lowering medications to account for individual preferences, which would increase treatment adherence and improve glycaemic control.

Both oral and subcutaneous formulations of GLP-1 receptor agonists have been investigated in clinical trials in individuals with type 2 diabetes, overweight, or obesity.[5,6] Oral semaglutide showed superior glycaemic control compared with other oral glucose-lowering treatments, showed clinically meaningful weight reduction, and had a well established safety profile.[6–10] Additionally, oral semaglutide was associated with a reduced risk of major adverse cardiovascular events compared with placebo.[11] Another GLP-1 receptor agonist, orforglipron, has been approved by the US Food and Drug Administration for weight management and is in development for the treatment of people with type 2 diabetes. It is a small-molecule, non-peptide oral GLP-1 receptor agonist that has shown improvements in both glycated haemoglobin (HbA_{1c}) and bodyweight compared with placebo in phase 3 trials.[12–14]

There is accumulating evidence that treatment regimens that use a combination of drugs with differing but complementary mechanisms of action for weight reduction, glycaemic control, and other cardiometabolic parameters can result in better outcomes than monotherapy.[15] GLP-1 receptor agonists in combination with amylin analogues are being investigated for the treatment of type 2 diabetes and overweight or obesity. Amylin is co-secreted with insulin from pancreatic β cells in response to food consumption. Amylin analogues mimic amylin and, similar to GLP-1 receptor agonists, contribute to weight management and glycaemic control, as shown in clinical trials in individuals with type 2 diabetes and obesity.[16,17] Furthermore, a combination therapy comprising a fixed-dose combination of cagrilintide and semaglutide (CagriSema, subcutaneous formulation) showed superior reductions in HbA_{1c} and bodyweight compared with the monocomponents in individuals with type 2 diabetes.[15]

Zenagamtide (formerly known as amycretin) is a unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors. Zenagamtide is being investigated for the treatment of overweight, obesity, and type 2 diabetes.[18,19] To date, both oral and subcutaneous formulations of zenagamtide have been investigated in individuals living with overweight or obesity without type 2 diabetes. To enhance oral absorption, the zenagamtide peptide was co-formulated with the small fatty-acid derivative sodium N-(8-[2-hydroxylbenzoyl] amino) caprylate (SNAC).[19] A first-in-human phase 1 trial investigating the safety and tolerability of oral zenagamtide in adults with overweight or obesity without type 2 diabetes reported greater reductions in bodyweight and HbA_{1c} than with placebo.[19] Additionally, in a phase 1b/2a trial also in adults with overweight or obesity without type 2 diabetes, once-weekly subcutaneous zenagamtide treatment resulted in greater weight loss than placebo.[18] In both trials, the safety and tolerability profile for zenagamtide was consistent with that reported for GLP-1 and amylin receptor agonists.[18,19]

The current dose-finding phase 2 trial is the first clinical evaluation of zenagamtide in individuals with type 2 diabetes. We aimed to investigate the dose–response relationship with respect to the efficacy and safety of once-daily oral zenagamtide compared with placebo in adults with type 2 diabetes.

Methods

Study design and participants

This multicentre, randomised, parallel, double-blind (within treatment groups), placebo-controlled, dose-finding, phase 2 trial assessed the dose–response relationship of three once-daily oral doses of zenagamtide (6, 25, and 50 mg) and compared their effects on glycaemic control and bodyweight with placebo. The trial was conducted at 83 hospital and clinic sites across 11 countries (Bulgaria, Croatia, Germany, Greece, Hungary, Japan, Poland, Romania, Slovakia, Spain, and the USA). The trial consisted of a 3-week screening period, a 36-week intervention period, and a 4-week follow-up period (appendix p 14). A summary of the study protocol and the statistical analysis plan can be found in the appendix (pp 22–201).

See Online for appendix

Adults aged 18–75 years with type 2 diabetes (HbA_{1c} 7·0–10·0% [53–86 mmol/mol]), treated with stable doses of metformin with or without a SGLT2 inhibitor, and with a BMI of 23·0 to <50·0 kg/m² were eligible for enrolment. Participants with abnormal calcium metabolism or thyroid–calcitonin-related cancer were excluded. Per protocol, a maximum of 10% of randomly assigned participants could have a BMI lower than 25·0 kg/m². Full inclusion and exclusion criteria are detailed in the appendix (appendix p 4–5). Participants were asked to report their biological sex (male or female) by trial personnel; their responses were collected in an electronic data capture system.

The trial was conducted in compliance with the principles of the Declaration of Helsinki and in accordance with the Good Clinical Practice guidelines of the International Conference for Harmonisation.[20,21] The protocol, consent form, and other relevant documents were reviewed and approved by the appropriate institutional review boards or independent ethics committees in each country (appendix p 6). All participants provided written informed consent before trial entry. There was no participant involvement in the design, conduct, or reporting of the trial. This trial was registered with ClinicalTrials.gov (NCT06542874) on Aug 5, 2024, and has been completed.

Randomisation and masking

A randomisation and trial supplies management system was used to assign eligible participants (7:5) to the once-weekly subcutaneous zenagamtide or once-daily oral zenagamtide treatment groups. Details of further randomisation for the once-weekly subcutaneous treatment group are reported separately.[22] Those assigned to the oral treatment group were randomly assigned (5:1) to oral zenagamtide or placebo. Participants assigned to oral zenagamtide were then randomly assigned (1:1:1) to one of three oral zenagamtide treatment groups with different maintenance doses (6, 25, and 50 mg). Participants assigned to placebo were similarly allocated to matched placebo regimens corresponding to each dosing schedule; individual placebo groups were pooled for analysis. This strategy planned for 50 participants to receive zenagamtide and ten to receive placebo in each of the groups. Participants were assigned to the next available treatment according to the randomisation schedule, and randomisation was stratified by HbA_{1c} (<8·5% or ≥8·5%) at screening and country of residence (Japan or other countries). The overall trial population was aimed to include approximately 10% participants from Japan to support their inclusion in phase 3 trials. The trial was double-blind within each dose level, with participants and investigators masked to allocation to zenagamtide or matched placebo. Participants received one tablet a day with varying zenagamtide doses depending on which group the participant was assigned to in the dosing scheme. Corresponding tablets not containing zenagamtide were provided to participants in the placebo group. Participants and investigators were not masked to the assigned dose level. To maintain masking of investigators, site staff, participants, and the sponsor throughout the trial, the zenagamtide oral formulation and placebo were visually identical and provided in identical packaging.

Procedures

Trial treatment (oral zenagamtide or placebo) was administered once daily. Similar to other SNAC-based oral medications,[6] the medication was taken in the morning on an empty stomach, 30 min before food, liquids, or other oral medications (additional details are provided in the appendix p 3). The starting dose of oral zenagamtide was 1·5 mg, with dose escalations every 4 weeks until the maintenance dose was reached (6, 25, or 50 mg). The starting and escalation doses of zenagamtide were chosen on the basis of the safety, tolerability, and pharmacokinetics observed in the phase 1 first-in-human dose study with oral zenagamtide (NCT05369390).[19] Depending on the maintenance dose level, the maintenance period was 16, 20, or 28 weeks, up to a total treatment period of 36 weeks (appendix p 14). Dose modification from the randomly assigned dose was not allowed within the fixed-dose-escalation trial design. Therefore, if the planned dose was not tolerated, the participant permanently discontinued trial treatment. Randomly assigned treatments were continued until week 36 or until the participant permanently discontinued treatment. In cases of persistent hyperglycaemia, rescue medication (intensification or initiation of glucose-lowering medications, excluding GLP-1 receptor agonists, DPP-4 inhibitors, GLP-1 and GIP receptor dual agonists, and amylin analogues) was prescribed to participants at the investigators' discretion (criteria provided in the appendix p 3). Blood glucose concentrations were measured using the provided blood glucose meter (Abbott FreeStyle Optium Neo, Abbott, Berkshire, UK) when a hypoglycaemic episode was suspected. Participants were provided with a continuous glucose monitoring (CGM) device (Dexcom G6, Dexcom, San Diego, CA, USA). CGM data were collected during three prespecified measurement periods of approximately 10 days starting at screening (used as baseline), week 18, and week 34. Both the participants and investigators were masked to CGM data, and these data were not used for dose adjustments or reporting of hypoglycaemic events.

Outcomes

The primary outcome was the change in HbA_{1c} from baseline to week 36. Supportive (ie, to reinforce or provide context for the primary outcome) secondary outcomes were changes from baseline to week 36 in: relative and absolute bodyweight; fasting plasma glucose (FPG); BMI; waist circumference; systolic blood pressure (SBP) and 24 h SBP; high-sensitivity C-reactive protein (hsCRP); total cholesterol, HDL cholesterol, LDL cholesterol, and triglycerides; urinary albumin-to-creatinine ratio (UACR); and estimated glomerular filtration rate (eGFR) based on the Chronic Kidney Disease Epidemiology Collaboration (Creatinine-Cystatin C) equation (2021).[23] Other secondary outcomes included change in percentage of time in range (TIR; interstitial glucose 3·9–10·0 mmol/L [70–180 mg/dL]) from baseline to weeks 20 and 36; the number of participants without macroalbuminuria (UACR <300 mg/g) at baseline developing new-onset macroalbuminuria (UACR ≥300 mg/g) at week 36; and the number of adverse events from baseline (week 0) to end of trial (week 40). Prespecified exploratory outcomes included attainment of HbA_{1c} targets (<7·0%, ≤6·5%, and <5·7%) and weight loss targets (≥5%, ≥10%, ≥15%, ≥20%, and ≥25% weight reduction) at week 36, and change in pulse from baseline to week 36. Further outcomes reported alongside the prespecified study outcomes included the percentage of TIR, time in tight range (TITR; 3·9–7·8 mmol/L [70–140 mg/dL]), and time below range (TBR; <3·9 mmol/L [70 mg/dL] and <3·0 mmol/L [54 mg/dL]).

Statistical analyses

The primary objective of this trial was to show and characterise the dose–response relationship of once-daily oral zenagamtide, and to compare the effect of varying doses with that of placebo, for change in HbA_{1c} from baseline to week 36. Sample size calculations were based on simulations using a maximum effect (E_{max}) model for the primary outcome, assuming that the baseline effect was 0, the median effective dose (ED_{50}) was 5 mg for oral zenagamtide, E_{max} was –2·2%, and that the SD was 1·1%. The SD in this trial was based on results from a previous trial with CagriSema (cagrilintide 2·4 mg and semaglutide 2·4 mg) in individuals with type 2 diabetes, in which the SD was 0·9% after 32 weeks of treatment.[24] The E_{max} and ED_{50} assumptions were based on a phase 1 trial of zenagamtide.[19] Based on these assumptions, it was considered that the random assignment of 50 participants to each active group would ensure sufficient precision to characterise a dose–response relationship for HbA_{1c}.

The full analysis set included all randomly assigned participants, whereas the safety analysis set included all randomly assigned participants exposed to treatment. The efficacy estimand was estimated based on the full analysis set using data collected while participants remained on trial treatment without initiation of rescue medication (appendix p 7). Imputation of missing data was handled by multiple imputation, assuming that missing data were missing at random.

An ANCOVA model with randomly assigned treatment and HbA_{1c} stratification (<8·5% *vs* ≥8·5%) as fixed factors and baseline HbA_{1c} as a fixed covariate was used to compare the effect of varying zenagamtide doses with placebo for change in HbA_{1c} from baseline to week 36. A similar approach was used to evaluate the relative change in bodyweight, with baseline bodyweight used as a fixed covariate. Change in BMI, eGFR, FPG, hsCRP, lipid parameters, average 24 h SBP, SBP, TIR, UACR, and waist circumference from baseline to week 36 were analysed using an ANCOVA model with the outcome as a dependent variable, randomly assigned treatment and strata as fixed factors, and the baseline assessment corresponding to the outcome as a fixed covariate. Attainment of HbA_{1c} and weight loss targets at week 36 were analysed using a logistic regression model with the outcome as a dependent variable, randomised treatment and strata as fixed factors, and the baseline assessment corresponding to the outcome as a fixed covariate. All statistical analyses were prespecified. p values were nominal and not adjusted for multiplicity, as such, the results should be interpreted as exploratory or descriptive. The number of adverse events were summarised descriptively by treatment group. Imputation was

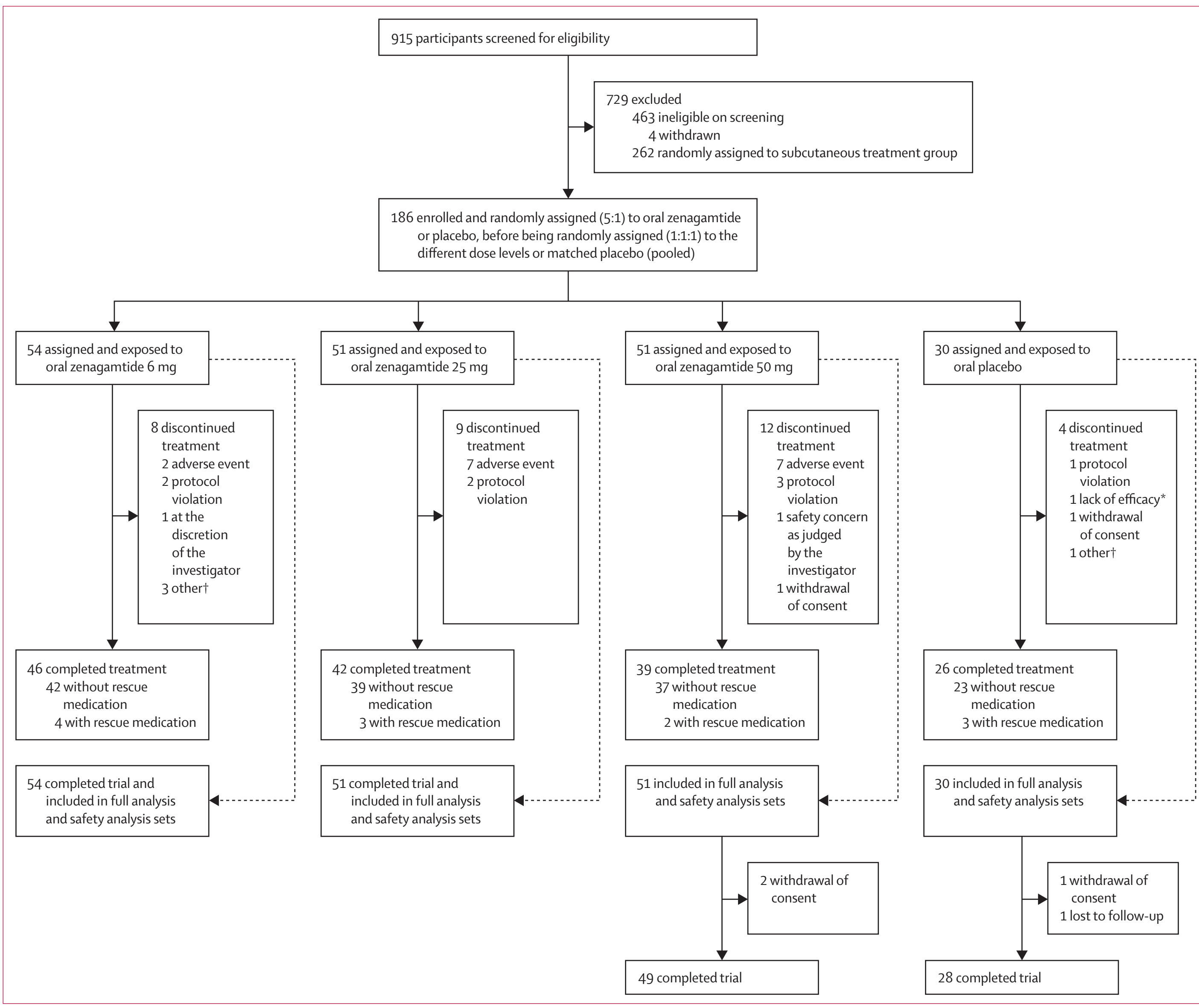

Figure 1: **Trial profile**

Details of screening failures are reported in the appendix (p 8). Treatment completers were defined as participants who completed treatment with the trial product according to the end-of-investigational medicinal product treatment form. *Lack of efficacy was an option that the investigator could choose in the electronic case report form; no additional details available. †As reported by the investigator, other reasons for discontinuation included missed doses as per protocol (one participant receiving 6 mg), participant's last dose was taken >9 days ago (one participant receiving 6 mg), participant decided to discontinue treatment following adverse events (one participant receiving 6 mg), and personal reasons (one participant receiving placebo).

	Zenagamtide 6 mg (n=54)	Zenagamtide 25 mg (n=51)	Zenagamtide 50 mg (n=51)	Pooled placebo (n=30)
Sex				
Male	33 (61%)	27 (53%)	32 (63%)	21 (70%)
Female	21 (39%)	24 (47%)	19 (37%)	9 (30%)
Age, years	54·9 (9·8)	58·0 (8·6)	54·4 (9·7)	55·1 (10·8)
HbA_{1c}, %	8·1 (0·9)	8·0 (0·9)	7·9 (0·7)	8·0 (0·8)
HbA_{1c}, mmol/mol	65·1 (9·5)	64·1 (9·6)	62·6 (8·0)	64·3 (8·3)
HbA_{1c} strata				
<8·5%	36 (67%)	36 (71%)	40 (78%)	22 (73%)
≥8·5%	18 (33%)	15 (29%)	11 (22%)	8 (27%)
FPG, mmol/L	9·5 (3·7)	8·8 (2·2)	8·9 (2·5)	8·7 (2·4)
FPG, mg/dL	171·2 (66·0)	157·9 (39·2)	160·7 (44·2)	156·1 (42·9)
Diabetes duration, years	7 (6)	7 (4)	8 (7)	10 (7)
Bodyweight, kg	100·7 (21·7)	99·5 (23·1)	104·1 (21·6)	99·7 (18·4)
BMI, kg/m^2	34·6 (6·1)	35·0 (6·5)	35·0 (6·4)	34·5 (6·1)
Waist circumference, cm	113·8 (16·1)	111·0 (15·6)	114·7 (14·9)	112·9 (13·4)
Systolic blood pressure, mm Hg	131·6 (13·0)	132·6 (14·2)	129·7 (12·2)	128·5 (11·2)
Diastolic blood pressure, mm Hg	81·4 (9·6)	81·3 (7·9)	80·5 (8·5)	79·7 (8·1)
eGFR,* mL/min per 1·73 m²	97 (20)	97 (16)	98 (17)	101 (18)
SGLT2 inhibitor use at randomisation	16 (30%)	23 (45%)	21 (41%)	9 (30%)
Race				
White	42 (78%)	44 (86%)	42 (82%)	24 (80%)
Asian	10 (19%)	7 (14%)	8 (16%)	5 (17%)
Black or African American	1 (2%)	0	1 (2%)	1 (3%)
Native Hawaiian or other Pacific Islander	1 (2%)	0	0	0

Data are n (%) or mean (SD). The full analysis set included all randomly assigned participants. CKD-EPI=Chronic Kidney Disease Epidemiology Collaboration. eGFR=estimated glomerular filtration rate. FPG=fasting plasma glucose. *eGFR calculated using the CKD-EPI (Creatinine-Cystatin C) equation (2021).[23]

Table 1: **Participant demographics and baseline characteristics (full analysis set)**

performed using SAS (version 9.4) and statistical analyses were performed using R (version 4.0.4).

Role of the funding source

Representatives of Novo Nordisk were involved in the trial design and conduct; data collection, management, analysis, and interpretation; and preparation, review, and approval of the manuscript. Novo Nordisk did not have the right to veto publication or to control the decision regarding journal submission. Additionally, Novo Nordisk funded medical writing support to aid with the drafting of the manuscript under the direction of the authors.

Results

Participants were recruited between Aug 7 and Dec 27, 2024. Of the 915 individuals who were screened, 262 were randomly assigned to the subcutaneous treatment group and 186 were randomly assigned to the oral treatment group. Details of screening failures are reported in the appendix (p 8). Details of the subcutaneous treatment group are reported separately.[22] The 186 participants assigned to the oral treatment group were randomly assigned to receive oral zenagamtide (n=156) or placebo (n=30) and were all exposed to treatment. Participants assigned to oral zenagamtide were assigned to one of three maintenance doses, zenagamtide 6 mg (n=54), zenagamtide 25 mg (n=51), and zenagamtide 50 mg (n=51). Participants assigned to placebo were allocated to matched placebo regimens and pooled for analysis (figure 1). Baseline characteristics were generally similar between treatment groups (table 1). At baseline, participants had a mean age of 55·6 years (SD 9·7), HbA_{1c} of 8·0% (0·8), BMI of 34·8 kg/m² (6·3), and diabetes duration of 8 years (6). 113 (61%) of 186 participants were male. Overall, 153 (82%) of 186 participants completed trial treatment; eight (15%) of 54 participants in the 6 mg group, nine (18%) of 51 participants in the 25 mg group, and 12 (24%) of 51 participants in the 50 mg group discontinued treatment. In the placebo group, four (13%) of 30 participants discontinued treatment. Most treatment discontinuations were due to adverse events (16 [9%] of 186 participants). The largest numbers of discontinuations due to adverse events were reported in the 25 mg (seven participants) and 50 mg (seven participants) treatment groups (figure 1). In the 25 mg and 50 mg treatment groups, discontinuations primarily occurred during the initial weeks after reaching the maintenance dose (appendix p 15). Rescue medication was required by four (7%) of 54 participants in the 6 mg treatment group, three (6%) of 51 participants in the 25 mg treatment group, two (4%) of 51 participants in the 50 mg treatment group, and three (10%) of 30 participants in the placebo group (figure 1).

Unless otherwise stated, efficacy results are estimated values based on the efficacy estimand. Zenagamtide showed a significantly greater change in HbA_{1c} from baseline to week 36 for all doses compared with placebo (table 2; figure 2A), with a significant dose–response relationship (appendix p 10). From a baseline range of 7·9–8·1%, the estimated change in HbA_{1c} at week 36 was –0·9% with zenagamtide 6 mg (estimated treatment difference [ETD] *vs* placebo: –0·5% [95% CI –1·04 to –0·04]; p=0·033), –1·3% with zenagamtide 25 mg (–0·99% [–1·49 to –0·49]; p=0·0001), and –1·4% with zenagamtide 50 mg (–1·09% [–1·59 to –0·59]; p<0·0001; table 2). A higher proportion of participants attained HbA_{1c} targets (<7%, ≤6·5%, and <5·7%) at week 36 with all zenagamtide doses than with placebo (figure 2B).

From baseline to week 20, the observed percentage of TIR (3·9–10·0 mmol/L [70–180 mg/dL]) increased with all zenagamtide doses, as did the percentage of time in tight range (TITR; 3·9–7·8 mmol/L [70–140 mg/dL]; table 2; appendix p 16). The observed mean percentage of TIR was 73·8–82·9% (equivalent to approximately 17 h 43 min to 20 h per day) at week 36 for all the zenagamtide doses investigated (figure 2C). There was a significantly greater increase in percentage of TIR from baseline to week 36 with zenagamtide 25 mg and 50 mg than with placebo (table 2). The estimated change in percentage of TIR from baseline to week 36 ranged from 16·7% with zenagamtide 6 mg (ETD *vs* placebo 12·9% [95% CI –3·61 to 29·39]; p=0·12) to 28·2% with zenagamtide 50 mg (24·3% [8·70 to 39·94]; p=0·0026; table 2). In the placebo group, the estimated change in percentage of TIR from baseline to week 36 was 3·8%. From baseline to week 36, the observed percentage of TITR increased with all zenagamtide doses (figure 2C). The observed percentage of time below range (TBR) below 3·9 mmol/L (70 mg/dL) and TBR below 3·0 mmol/L (54 mg/dL) ranged from 0·3% to 0·6% and from 0·1% to 0·3%, respectively, for all zenagamtide doses at week 36 (figure 2C).

	Zenagamtide 6 mg (n=54)	Zenagamtide 25 mg (n=51)	Zenagamtide 50 mg (n=51)	Pooled placebo (n=30)
HbA_{1c}, %				
Baseline	8·1 (0·9)	8·0 (0·9)	7·9 (0·7)	8·0 (0·8)
Change from baseline at week 36	−0·9	−1·3	−1·4	−0·4
ETD *vs* placebo (95% CI); p value	−0·54 (−1·04 to −0·04); 0·033	−0·99 (−1·49 to −0·49); 0·0001	−1·09 (−1·59 to −0·59); <0·0001	··
TIR 3·9–10·0 mmol/L (70–180 mg/dL), %				
Baseline	50·6 (31·2)	56·4 (28·3)	55·2 (28·1)	55·6 (22·5)
Change from baseline at week 20	22·8	28·0	32·9	1·5
ETD *vs* placebo (95% CI); p value	21·2 (10·22 to 32·25); 0·0002	26·5 (15·62 to 37·37); <0·0001	31·4 (20·62 to 42·13); <0·0001	··
Change from baseline at week 36	16·7	22·2	28·2	3·8
ETD *vs* placebo (95% CI); p value	12·9 (−3·61 to 29·39); 0·12	18·4 (2·68 to 34·12); 0·022	24·3 (8·70 to 39·94); 0·0026	··
FPG, mmol/L				
Baseline	9·5 (3·7)	8·8 (2·2)	8·9 (2·5)	8·7 (2·4)
Change from baseline at week 36	−1·3	−2·1	−2·3	−0·4
ETD *vs* placebo (95% CI); p value	−0·87 (−1·97 to 0·23); 0·12	−1·66 (−2·71 to −0·60); 0·0023	−1·93 (−2·99 to −0·87); 0·0005	··
FPG, mg/dL				
Baseline	171·2 (66·0)	157·9 (39·2)	160·7 (44·2)	156·1 (42·9)
Change from baseline at week 36	−23·3	−37·4	−42·3	−7·5
ETD *vs* placebo (95% CI); p value	−15·76 (−35·59 to 4·07); 0·12	−29·88 (−48·90 to −10·85); 0·0023	−34·77 (−53·94 to −15·60); 0·0005	··
Bodyweight change, %				
Baseline, kg	100·7 (21·7)	99·5 (23·1)	104·1 (21·6)	99·7 (18·4)
Change from baseline at week 36	−5·0	−10·3	−9·0	−3·0
ETD *vs* placebo (95% CI); p value	−1·91 (−5·01 to 1·20); 0·23	−7·29 (−10·44 to −4·15); <0·0001	−5·92 (−9·08 to −2·77); 0·0003	··
Bodyweight, kg				
Baseline	100·7 (21·7)	99·5 (23·1)	104·1 (21·6)	99·7 (18·4)
Change from baseline at week 36	−5·1	−10·1	−8·8	−3·3
ETD *vs* placebo (95% CI); p value	−1·77 (−4·88 to 1·33); 0·26	−6·82 (−9·97 to −3·66); <0·0001	−5·45 (−8·63 to −2·28); 0·0009	··
BMI, kg/m^2				
Baseline	34·6 (6·1)	35·0 (6·5)	35·0 (6·4)	34·5 (6·1)
Change from baseline at week 36	−1·8	−3·6	−3·0	−1·2
ETD *vs* placebo (95% CI); p value	−0·6 (−1·72 to 0·45); 0·25	−2·4 (−3·51 to −1·30); <0·0001	−1·8 (−2·95 to −0·73); 0·0013	··
Waist circumference, cm				
Baseline	113·8 (16·1)	111·0 (15·6)	114·7 (14·9)	112·9 (13·4)
Change from baseline at week 36	−3·5	−7·6	−5·9	−3·7
ETD *vs* placebo (95% CI); p value	0·25 (−3·00 to 3·51); 0·88	−3·84 (−7·22 to −0·46); 0·026	−2·18 (−5·47 to 1·11); 0·19	··
hsCRP, mg/L				
Baseline	5·1 (5·6)	7·1 (19·4)	5·4 (5·4)	4·4 (4·9)
Ratio to baseline at week 36	0·8	0·4	0·7	0·8
ETR *vs* placebo (95% CI); p value	1·00 (0·58 to 1·71); 0·99	0·55 (0·32 to 0·93); 0·027	0·85 (0·49 to 1·48); 0·57	··
HDL cholesterol, mmol/L				
Baseline	1·2 (0·2)	1·2 (0·2)	1·2 (0·3)	1·2 (0·3)
Ratio to baseline at week 36	1·0	1·1	1·0	1·1
ETR *vs* placebo (95% CI); p value	0·99 (0·91 to 1·07); 0·75	1·05 (0·97 to 1·14); 0·27	0·98 (0·91 to 1·06); 0·64	··

(Table 2 continues on next page)

	Zenagamtide 6 mg (n=54)	Zenagamtide 25 mg (n=51)	Zenagamtide 50 mg (n=51)	Pooled placebo (n=30)
(Continued from previous page)				
LDL cholesterol, mmol/L				
Baseline	2·6 (1·1)	2·3 (1·0)	2·6 (1·0)	2·9 (0·9)
Ratio to baseline at week 36	0·9	1·0	0·9	0·9
ETR *vs* placebo (95% CI); p value	0·99 (0·83 to 1·18); 0·87	1·09 (0·92 to 1·30); 0·33	1·02 (0·86 to 1·22); 0·83	..
Total cholesterol, mmol/L				
Baseline	4·7 (1·2)	4·3 (1·1)	4·7 (1·1)	5·0 (1·2)
Ratio to baseline at week 36	0·9	1·0	0·9	0·9
ETR *vs* placebo (95% CI); p value	0·97 (0·88 to 1·07); 0·58	1·03 (0·93 to 1·14); 0·55	0·98 (0·89 to 1·08); 0·70	..
Triglycerides, mmol/L				
Baseline	2·4 (1·5)	1·7 (0·7)	1·9 (0·8)	2·0 (1·1)
Ratio to baseline at week 36	0·9	0·8	0·9	1·0
ETR *vs* placebo (95% CI); p value	0·87 (0·70 to 1·08); 0·21	0·80 (0·65 to 1·00); 0·048	0·87 (0·70 to 1·09); 0·23	..
SBP, mm Hg				
Baseline	131·6 (13·1)	132·6 (14·2)	129·7 (12·2)	128·5 (11·2)
Change from baseline at week 36	−4·0	−8·5	−10·7	−3·1
ETD *vs* placebo (95% CI); p value	−0·92 (−8·01 to 6·17); 0·80	−5·42 (−12·80 to 1·96); 0·15	−7·56 (−14·69 to −0·43); 0·038	..
24 h SBP, mm Hg				
Baseline	126·8 (11·9)	132·4 (17·2)	128·2 (12·4)	127·0 (10·8)
Change from baseline at week 36	−2·8	−3·7	−7·0	−2·7
ETD *vs* placebo (95% CI); p value	−0·10 (−6·04 to 5·83); 0·97	−0·97 (−7·35 to 5·40); 0·76	−4·27 (−11·01 to 2·46); 0·21	..
eGFR,* mL/min per 1·73 m²				
Baseline	97 (20)	97 (16)	98 (17)	101 (18)
Change from baseline at week 36	2	4	5	0
ETD *vs* placebo (95% CI); p value	1·95 (−2·30 to 6·20); 0·37	3·90 (−0·53 to 8·32); 0·084	4·97 (0·66 to 9·28); 0·024	..
UACR, mg/g				
Baseline†	28·3 (199·8)	29·4 (148·1)	20·4 (126·8)	16·2 (140·6)
Ratio to baseline at week 36	0·7	0·7	0·6	0·7
ETR *vs* placebo (95% CI); p value	0·99 (0·52 to 1·88); 0·97	1·09 (0·56 to 2·10); 0·80	0·91 (0·45 to 1·86); 0·80	..

The full analysis set included all randomly assigned participants. Data at baseline are observed values and are shown as mean (SD), unless otherwise stated. All other data are estimated values based on the efficacy estimand, incorporating the full analysis set and the on-treatment without rescue medication datapoint set. CKD-EPI=Chronic Kidney Disease Epidemiology Collaboration. eGFR=estimated glomerular filtration rate. ETD=estimated treatment difference. ETR=estimated treatment ratio. FPG=fasting plasma glucose. hsCRP=high-sensitivity C-reactive protein. SBP=systolic blood pressure. TIR=time in range. UACR=urinary albumin-to-creatinine ratio. *eGFR calculated using the CKD-EPI (Creatinine-Cystatin C) equation (2021).[23] †Geometric mean (coefficient of variation).

Table 2: **Summary of key outcomes (full analysis set)**

FPG decreased from baseline to week 36 in all zenagamtide treatment groups (table 2). The estimated change in FPG from baseline to week 36 ranged from –1·3 mmol/L (–23·3 mg/dL) with zenagamtide 6 mg (ETD *vs* placebo –0·87 mmol/L [95% CI –1·97 to 0·23 mmol/L]; –15·76 mg/dL [–35·59 to 4·07 mg/dL]; p=0·12) to –2·3 mmol/L (–42·3 mg/dL) with zenagamtide 50 mg (–1·93 mmol/L [–2·99 to –0·87 mmol/L]; –34·77 mg/dL [–53·94 to –15·60 mg/dL]; p=0·0005). In the placebo group, FPG decreased by 0·4 mmol/L (7·5 mg/dL) over 36 weeks (table 2).

Bodyweight, BMI, and waist circumference decreased throughout the trial for all zenagamtide doses (appendix p 17). Statistically significant reductions in percentage change in bodyweight from baseline to week 36 were observed in the zenagamtide 25 mg and 50 mg groups compared with the placebo group (table 2). The estimated percentage change in bodyweight from baseline to week 36 was –5·0% with zenagamtide 6 mg (ETD *vs* placebo –1·91% [–5·01 to 1·20]; p=0·23), –10·3% with zenagamtide 25 mg (–7·29% [–10·44 to –4·15]; p<0·0001), and –9·0% with zenagamtide 50 mg (–5·92% [–9·08 to –2·77]; p=0·0003; table 2). In the placebo group, the estimated percentage change in bodyweight was –3·1% over 36 weeks. None of the zenagamtide doses showed a clear weight loss plateau at week 36 (figure 2D).

There were statistically significant decreases in bodyweight and BMI versus placebo for the zenagamtide 25 mg and 50 mg doses from baseline to week 36 (table 2). A statistically significant decrease in waist circumference compared with placebo was observed with zenagamtide 25 mg (table 2). A greater proportion of participants attained predefined weight loss targets

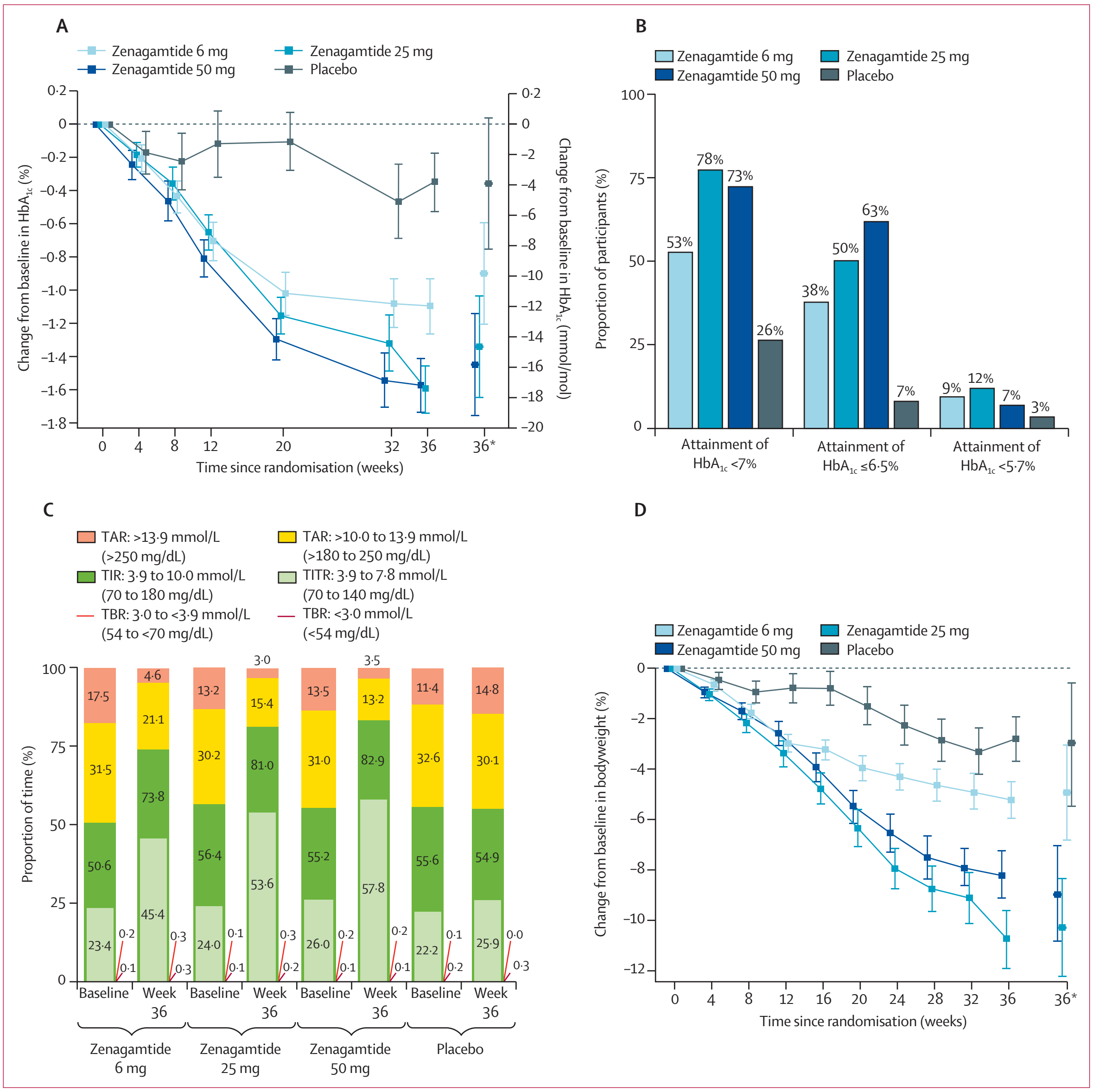

Figure 2: **Key efficacy outcomes**

Data from the on-treatment without rescue medication observation period. (A) Mean change in HbA_{1c} from baseline to week 36. Error bars show standard error of the mean. Estimated values from the primary analysis. (B) Proportion of participants who attained HbA_{1c} targets at week 36. Each binary responder outcome was analysed using a logistic regression model with the outcome as the dependent variable, with randomly assigned treatment and strata as fixed-effect factors, and with baseline of the underlying parameter as a fixed-effect covariate. Missing data were imputed using multiple imputation. (C) CGM metrics at baseline and week 36. Outcomes assessed as observed means. CGM assessments are averages calculated over a 10-day period. Time spent is defined as the proportion of time spent in a given range. (D) Mean change in bodyweight from baseline to week 36. Error bars show standard error of the mean. Estimated values from the primary analysis. CGM=continuous glucose monitoring. TAR=time above range. TBR=time below range. TIR=time in range. TITR=time in tight range. *Data shown at week 36 are the estimated mean values and the corresponding standard error from statistical analysis.

(≥5%, ≥10%, ≥15%, ≥20%, and ≥25% weight reduction) with zenagamtide than with placebo at week 36 (appendix p 11).

For SBP, there was a significant decrease from baseline to week 36 of –10·7 mm Hg (ETD *vs* placebo –7·56 [95% CI –14·69 to –0·43]; p=0·038) with zenagamtide 50 mg; however, there were no significant differences with any zenagamtide dose versus placebo in change in average 24 h SBP from baseline to week 36 (table 2). At week 36, the change from baseline in pulse ranged from 1·96 bpm to 4·84 bpm in the zenagamtide groups compared with 0·61 bpm in the placebo group (appendix pp 12, 18). The increases in pulse rate occurred primarily during dose escalation (appendix p 18). No significant increase was observed for any of the zenagamtide groups compared with placebo (appendix p 12).

	Zenagamtide 6 mg (n=54)	Zenagamtide 25 mg (n=51)	Zenagamtide 50 mg (n=51)	Pooled placebo (n=30)
Adverse events				
Total adverse events	36 (67%)	39 (77%)	36 (71%)	20 (67%)
Serious adverse events	2 (4%)	2 (4%)	3 (6%)	0
Severe adverse events	1 (2%)	1 (2%)	3 (6%)	0
Moderate adverse events	10 (19%)	15 (29%)	19 (37%)	4 (13%)
Mild adverse events	33 (61%)	36 (71%)	29 (57%)	18 (60%)
Adverse events probably related to trial product	9 (17%)	16 (31%)	17 (33%)	6 (20%)
Adverse events possibly related to trial product	14 (26%)	16 (31%)	13 (26%)	2 (7%)
Outcome: not recovered*	10 (19%)	17 (33%)	12 (24%)	7 (23%)
Outcome: fatal	0	0	0	0
Action taken: trial treatment discontinued	2 (4%)	7 (14%)	7 (14%)	0
Adverse events reported by ≥5% of participants†				
Nausea	6 (11%)	17 (33%)	18 (35%)	1 (3%)
Vomiting	4 (7%)	5 (10%)	8 (16%)	0
Diarrhoea	4 (7%)	4 (8%)	3 (6%)	2 (7%)
Constipation	4 (7%)	3 (6%)	2 (4%)	1 (3%)
Gastro-oesophageal reflux disease	2 (4%)	5 (10%)	2 (4%)	1 (3%)
Dyspepsia	2 (4%)	3 (6%)	0	0
Eructation	1 (2%)	3 (6%)	0	0
Frequent bowel movements	0	0	0	2 (7%)
Nasopharyngitis	5 (9%)	6 (12%)	4 (8%)	1 (3%)
Influenza	5 (9%)	0	1 (2%)	0
CRP increased	3 (6%)	0	2 (4%)	1 (3%)
Headache	1 (2%)	0	1 (2%)	2 (7%)
Decreased appetite	1 (2%)	6 (12%)	4 (8%)	0
Dyslipidaemia	1 (2%)	0	0	2 (7%)
Fatigue	3 (6%)	1 (2%)	4 (8%)	1 (3%)
Asthenia	1 (2%)	2 (4%)	3 (6%)	0
Back pain	0	3 (6%)	1 (2%)	1 (3%)
Hypotension	0	0	0	2 (7%)
Adverse events of special interest‡				
Gastrointestinal§	14 (26%)	21 (41%)	24 (47%)	7 (23%)
Cardiovascular disorders	2 (4%)	4 (8%)	4 (8%)	3 (10%)
Allergic reactions§	3 (6%)	0	0	2 (7%)
Neoplasms	1 (2%)	2 (4%)	4 (8%)	0
Drug-related hepatic disorders§	0	2 (4%)	3 (6%)	..
Retinal disorders	2 (4%)	4 (8%)	2 (4%)	1 (3%)
Rare events	0	0	2 (4%)	..
Dysaesthesia§	0	0	1 (2%)	0
Malignant tumours§	0	1 (2%)	0	0
Medication errors, abuse, and misuse§	1 (2%)	1 (2%)	0	1 (3%)
Pancreatitis§	1 (2%)	0	0	0
Acute gallstone disease§	0	0	0	0
Hypoglycaemia§¶	0	0	0	0

Data are the number of participants with an event (%). The safety analysis set included all randomly assigned participants. CRP=C-reactive protein. MedDRA=Medical Dictionary for Regulatory Activities. *Not recovered by the data cutoff date of Nov 3, 2025. †In any treatment group. ‡Evaluated using prespecified standardised MedDRA search queries or customised clusters of MedDRA preferred terms. §On-treatment adverse events (ie, those occurring from the first dose until 29 days after the last dose or last study site contact). ¶Severe hypoglycaemia (level 3).

Table 3: **Adverse events from baseline to week 40 (safety analysis set)**

A statistically significant improvement in hsCRP was observed from baseline to week 36 with zenagamtide 25 mg versus placebo (table 2). At week 36, there were no significant differences for HDL cholesterol, LDL cholesterol, or total cholesterol with any zenagamtide dose versus placebo (table 2; appendix p 19). A significant improvement compared with placebo was observed for triglycerides with zenagamtide 25 mg at week 36 (table 2). From baseline to week 36, there was a significant increase in eGFR with zenagamtide 50 mg compared with placebo (table 2). There were no significant differences compared with placebo in UACR for any zenagamtide dose (table 2). At week 36, one (8%) participant in the 6 mg zenagamtide group, one (8%) participant in the 25 mg zenagamtide group, and two (12%) participants in the 50 mg zenagamtide group had new onset macroalbuminuria (UACR ≥300 mg/g). In the placebo group, one (9%) participant had new onset macroalbuminuria (UACR ≥300 mg/g) at week 36.

The incidence of adverse events throughout the trial is reported in table 3. Most adverse events were mild to moderate in severity, probably or possibly related to trial treatment, and were resolved by data cutoff. Adverse events leading to permanent treatment withdrawal occurred in two (4%) of 54 participants in the zenagamtide 6 mg group, seven (14%) of 51 participants in the zenagamtide 25 mg group, and seven (14%) of 51 participants in the zenagamtide 50 mg group (table 3). No participants permanently discontinued treatment due to adverse events in the placebo group. The most frequently reported adverse events were gastrointestinal in nature (nausea, vomiting, and diarrhoea) and were mostly mild and moderate in severity (appendix p 20). Treatment discontinuation due to gastrointestinal adverse events was more frequently reported in the 25 mg (five [10%] of 51 participants) and 50 mg (six [12%] of 51 participants) dose groups than in the 6 mg group (two [4%] of 54 participants; appendix p 13). Treatment interruptions due to gastrointestinal adverse events occurred in one (2%), two (4%), and three (6%) participants in the 6 mg, 20 mg and 50 mg dose groups, respectively. The time to onset of gastrointestinal adverse events is shown in the appendix (p 21). The time to onset of the first event showed that there was a higher frequency of gastrointestinal adverse events in the 50 mg group than in the other treatment groups from approximately week 20 (when the 50 mg dose was initiated), reaching a plateau at week 34. From week 4 to week 40, the cumulative mean number of gastrointestinal adverse events was higher in the 25 mg group than in the other zenagamtide and placebo groups. Few events were reported from week 28 onwards (appendix p 21). Gastrointestinal adverse events primarily occurred during the dose-escalation period (appendix p 21).

Serious adverse events were reported in seven (4%) of 186 participants receiving zenagamtide:

two in the 6 mg group (one participant with adverse events of abdominal pain and acute pancreatitis [appendix p 3], and one participant with alcohol poisoning), two in the 25 mg group (coronary artery stenosis and umbilical hernia repair), and three in the 50 mg group (two adverse events of appendicitis and one of acute kidney injury). No serious adverse events were reported in the placebo group. No deaths occurred during the trial (table 3).

Discussion

To our knowledge, this was the first clinical evaluation of oral zenagamtide in individuals with type 2 diabetes. In this phase 2 trial, treatment with once-daily oral zenagamtide resulted in clinically meaningful improvements in glycaemic parameters (HbA_{1c} and TIR) and bodyweight, with significant reductions in the zenagamtide 25 mg and 50 mg groups compared with the placebo group from baseline to week 36. The safety and tolerability profile was consistent overall with the known safety profile of GLP-1 and amylin receptor agonists.[16,25]

Significant reductions in HbA_{1c} were observed for all the zenagamtide groups compared with the placebo group. Because mean baseline HbA_{1c} was 8·0% and 69 (37%) of 186 participants were already receiving a SGLT2 inhibitor at baseline, the estimated reductions in HbA_{1c} indicate that zenagamtide had clinically meaningful glycaemic efficacy in this phase 2 trial, with levels in the 6·5% range. In phase 3 trials in individuals with type 2 diabetes, selective oral GLP-1 receptor agonists have shown HbA_{1c} reductions of up to 2·2% but baseline HbA_{1c} was considerably higher.[6,12,26] It is important to note that indirect comparisons with previously published trials should be interpreted with caution due to differences in study designs, including study size and duration, and differences between the molecules, with zenagamtide being a unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors. Larger phase 3 trials are needed to support these findings and evaluate the long-term efficacy and safety of oral zenagamtide. Furthermore, in the current trial, the observed maximum reduction in HbA_{1c} was lower than predicted, which could reflect the relatively modest mean baseline HbA_{1c} (8.0%) and small sample size. Nevertheless, these findings provide useful context for refining effect size assumptions in the design of a subsequent phase 3 trial.

In the current trial, there were also significant and clinically meaningful improvements in the percentage of TIR from baseline to week 20 for all zenagamtide groups compared with placebo, and with zenagamtide 25 mg and 50 mg from baseline to week 36. Overall, up to an extra 6 h per day (24·3%) were spent within range after 36 weeks of treatment with zenagamtide 50 mg compared with placebo. At week 36, for all zenagamtide doses, the observed percentage of TIR was above the internationally recommended target of 70%.[27] The observed increases in TIR for the zenagamtide doses from baseline to week 36 exceeded the internationally recognised clinically meaningful threshold of 3% by approximately six to nine times.[27] Moreover, the percentage of TITR improved for all zenagamtide doses from baseline to week 36 without a corresponding increase in percentage of TBR. For all zenagamtide doses and placebo, the percentage of TBR below 3·9 mmol/L (<70 mg/dL) and below 3·0 mmol/L (<54 mg/dL) was below the CGM international consensus guideline targets of below 4% and below 1%, respectively, at week 36.[27]

Weight loss in individuals with type 2 diabetes is important because it leads to health benefits that extend beyond improvements in glycaemic control.[2,3,28] A bodyweight reduction of 5–10% leads to improvements in cardiovascular risk factors,[28] and a reduction of 10–15% can improve metabolic health and reduce the risk of long-term complications associated with type 2 diabetes.[2,29] In this study, the zenagamtide 25 mg and 50 mg doses resulted in clinically relevant and statistically significant reductions in bodyweight compared with placebo. In addition, none of the zenagamtide dose groups showed a clear weight loss plateau by week 36, suggesting further weight loss might be attainable with long-term treatment. Previously, in individuals with type 2 diabetes, mean percentage changes in bodyweight of –9·8% after 52 weeks and –9·2% after 36 weeks were reported with oral semaglutide 50 mg and oral orforglipron 36 mg, respectively.[6,30] In individuals with overweight or obesity without type 2 diabetes, mean percentage changes in bodyweight of –16·6% after 64 weeks and –12·4% after 72 weeks were observed with oral semaglutide 25 mg and oral orforglipron 36 mg, respectively.[14,31] The significant reductions in bodyweight observed with oral zenagamtide 25 mg and 50 mg in the present trial align with those previously seen in individuals with overweight or obesity without type 2 diabetes who received oral zenagamtide, in which a mean percentage change of up to –13·1% was observed with zenagamtide 2×50 mg (two tablets in a single daily dose) with no signs of plateau after 12 weeks of treatment.[19] The percentage change in bodyweight was lower in the present trial in individuals with type 2 diabetes compared with the previous trial in individuals with overweight or obesity without type 2 diabetes; this difference is expected, because several clinical trials have consistently shown that individuals with type 2 diabetes have less weight loss than those without type 2 diabetes.[32] Participants receiving placebo lost up to 3·0% of bodyweight from baseline to week 36. This reduction is not unexpected, because a holistic approach to the management of diabetes typically includes lifestyle and behavioural changes that promote weight loss, although we did not assess whether participants adopted these changes.[2]

Reductions in bodyweight, BMI, and waist circumference were numerically higher for participants receiving zenagamtide 25 mg than those receiving zenagamtide 50 mg. This finding could be partially explained by the higher proportion of female participants in the 25 mg treatment group than in the 50 mg group (47% *vs* 37%), because men tend to have less weight loss than women.[33]

Although treatment with zenagamtide 6 mg resulted in improvements in FPG, bodyweight, and BMI compared with placebo, these improvements were not statistically significant, indicating that higher doses might be required for significant improvements.

The increase in pulse rate observed in this trial occurred during dose escalation and remained relatively stable thereafter with no clear dose–response relationship. Increases in pulse rate have also been observed with other GLP-1-based treatments in people with type 2 diabetes.[18,34,35]

A significant increase in eGFR of 4·97 mL/min per 1·73 m² was observed for zenagamtide 50 mg with no significant change in UACR. However, the increase in eGFR should be interpreted with caution because eGFR was within the normal range at baseline (mean eGFR 97–101 mL/min per 1·73 m²). To better assess improvement in kidney function, zenagamtide should be investigated further in a study designed and powered to assess mechanisms for improvement in kidney function.

When interpreting data from this trial, the differences in the amount of time that participants spent at their respective maintenance dose should be considered. The zenagamtide 20 mg and 40 mg groups spent 20 weeks and 16 weeks at their maintenance dose, respectively, compared with 28 weeks for zenagamtide 6 mg and therefore did not have the same amount of time to affect HbA_{1c} and bodyweight. Additionally, there was a greater proportion of male participants than female participants enrolled in the trial (61% *vs* 39%), which could have attenuated the observed bodyweight loss.

In line with previously reported safety profiles for GLP-1 and amylin receptor agonists,[5,16] the most common adverse events were gastrointestinal, which were mostly mild or moderate in severity and related to the trial treatment. Gastrointestinal adverse events occurred primarily during the dose-escalation period, with a lower incidence once participants reached the maintenance dose. Treatment discontinuations due to adverse events ranged from 4% to 14% across the zenagamtide treatment groups, with more discontinuations reported with zenagamtide 25 mg and 50 mg. Most treatment discontinuations were related to gastrointestinal adverse events, primarily nausea and vomiting. Additionally, most treatment discontinuations occurred after escalation to the higher zenagamtide doses. In this trial, dose modification was prohibited; therefore, dose adjustments were not allowed to account for dose tolerability and participant preference, which might have led to higher discontinuation rates.

This trial had several strengths. A wide zenagamtide dose range (6–50 mg) was investigated to characterise the dose–response relationship and to inform dose selection for phase 3 trials. Both the 25 mg and 50 mg doses were effective, with an acceptable safety profile. Therefore, either dose would be suitable for investigation in a phase 3 trial, despite the absence of a clear dose–response relationship between the 25 mg and 50 mg doses. The absence of dose-dependent effects between the 25 mg and 50 mg doses could reflect the small population size, random variability, or differences between the treatment groups, and does not exclude the possibility of a true difference. The trial design incorporated double-blinding to placebo or zenagamtide within each dose level to minimise bias. Additionally, masked CGM data were collected to enable comprehensive assessment of glycaemic parameters.

This trial also had limitations that should be considered when interpreting results. First, the study duration was only 36 weeks; as such, the full glycaemic and bodyweight loss potential of oral zenagamtide in individuals with type 2 diabetes and the long-term safety profile could not be assessed, particularly because exposure to the higher zenagamtide doses was short compared with zenagamtide 6 mg (20 weeks and 16 weeks for zenagamtide 25 mg and 50 mg, respectively, *vs* 28 weeks for zenagamtide 6 mg). Second, treatment group sample sizes were small compared with those usually used in a phase 3 trial. Finally, although the trial was double-blind within each dose level to placebo or zenagamtide, participants were not masked to the dose level. Because participants were not masked to the dose level, treatment expectations and adverse event reporting could have been influenced, and the absence of masking might have contributed to anticipatory nocebo effects or discontinuation behaviour, especially among participants receiving higher doses. These factors should be considered when interpreting both tolerability and efficacy findings.

In conclusion, once-daily oral zenagamtide (25 mg and 50 mg doses) provided clinically meaningful and statistically significant improvements in glycaemic control and bodyweight compared with placebo in individuals with type 2 diabetes. In addition, no clear plateau in weight loss was observed at week 36 for any of the zenagamtide doses. The safety and tolerability profile of oral zenagamtide was consistent with that of other GLP-1 and amylin receptor agonists. To our knowledge, zenagamtide is the first unimolecular GLP-1, amylin, and calcitonin receptor agonist under investigation, with both oral and subcutaneous formulations in development to accommodate individual treatment preferences. Further mechanistic studies and long-term clinical follow-up are required to clarify how GLP-1, amylin, and calcitonin receptor agonism with zenagamtide differentially contribute to potential long-term efficacy-related benefits and safety. Overall, the findings reported in the current trial have clinically meaningful implications and support further investigation of once-daily oral zenagamtide in phase 3 trials.

Contributors

PM, MB, and JR were trial investigators and helped to obtain data. MK and A-LFC were responsible for medical oversight during the trial. LNV had access to the raw data and was responsible for the statistical considerations in the analysis and trial design. PM, MA, and MK accessed and verified the data. All authors participated in reviewing and interpreting the data outputs and in developing the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

PM has received honoraria as a consultant for Abbott, Biomea Fusion, Dexcom, Insulet, Minimed (Medtronic), and Novo Nordisk, and is a member of the speakers' bureau for Abbott, Dexcom, Insulet, Minimed (Medtronic), and Novo Nordisk. VRA reports institutional contracts with Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Corcept Therapeutics, Eli Lilly, Fractyl Health, Kailera Therapeutics, Novo Nordisk, Pfizer, Recordati, Rhythm Pharmaceuticals, and Servier, and has received consultancy fees from the Baim Institute for Clinical Research, Mediflix, Roche, and Sanofi. MB has received honoraria as a consultant and speaker from Abbott, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Daiichi–Sankyo, Lilly, MSD, Novo Nordisk, Novartis, Pfizer, and Sanofi, and reports chairing a clinical trial data safety monitoring board for Boehringer Ingelheim. MA, A-LFC, MK, and LNV are employees and shareholders of Novo Nordisk. JR reports clinical research grants from Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Corcept Therapeutics, Corxel, Eli Lilly, Hanmi Pharmaceutical, Merck, Novo Nordisk, Pfizer, Regeneron, Roche, Regor Therapeutics, Sanofi, Structure Therapeutics, and Terns Pharma; serves or has served on scientific advisory boards for, and has received honoraria or consulting fees from, Amgen, Applied Therapeutics, Arrowhead Pharmaceuticals, Biomea Fusion, Boehringer Ingelheim, Corcept, Eccogene, Eli Lilly, Hanmi Pharmaceutical, i2O Therapeutics, Novo Nordisk, Regeneron, Roche, Sanofi, Structure Therapeutics, and Terns Pharma; and has received honoraria for lectures from Eli Lilly, Novo Nordisk, and Sanofi.

Data sharing

An authorised researcher can request access to clinical trial data by submitting a research proposal for review and approval by Novo Nordisk and an internal independent review panel. Requests are usually considered after the research is finished and the main results have been published. If the research supports a regulatory application, requests will be considered after the product and its intended use are approved in both the EU and the USA. Participants' clinical data will be anonymised, following an approved internal process, before data are shared to external third parties. Details on how to request access to clinical data are online.

For more on **how to request access to clinical data** see novonordisk-trials.com

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References

1 Grant B, Sandelson M, Agyemang-Prempeh B, Zalin A. Managing obesity in people with type 2 diabetes. *Clin Med (Lond)* 2021; **21:** e327–31.

2 Davies MJ, Aroda VR, Collins BS, et al. Management of hyperglycemia in type 2 diabetes, 2022. A consensus report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care* 2022; **45:** 2753–86.

3 American Diabetes Association Professional Practice Committee for Diabetes*. 8. Obesity and weight management for the prevention and treatment of diabetes: Standards of care in diabetes–2026. *Diabetes Care* 2026; **49** (suppl 1)**:** S166–82.

4 Horii T, Masudo C, Takayanagi Y, Oikawa Y, Shimada A, Mihara K. Adherence and treatment discontinuation of oral semaglutide and once-weekly semaglutide injection at 12 month follow-up: Japanese real-world data. *J Diabetes Investig* 2024; **15:** 1578–84.

5 Wilding JPH, Batterham RL, Calanna S, et al, and the STEP 1 Study Group. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med* 2021; **384:** 989–1002.

6 Aroda VR, Aberle J, Bardtrum L, et al. Efficacy and safety of once-daily oral semaglutide 25 mg and 50 mg compared with 14 mg in adults with type 2 diabetes (PIONEER PLUS): a multicentre, randomised, phase 3b trial. *Lancet* 2023; **402:** 693–704.

7 Rodbard HW, Rosenstock J, Canani LH, et al, and the PIONEER 2 Investigators. Oral semaglutide versus empagliflozin in patients with type 2 diabetes uncontrolled on metformin: the PIONEER 2 trial. *Diabetes Care* 2019; **42:** 2272–81.

8 Xie Z, Gan B, Cao W, Liu J, Chen J. Semaglutide versus dulaglutide and liraglutide in Chinese patients with T2DM: a multicenter real-world study. *Obesity (Silver Spring)* 2026; **34:** 101–13.

9 Whorton A, Osman S, Johal J, Baig S, Jones AM, Jalal Z. Prescribing patterns and adverse effects of semaglutide: a real-world comparative evaluation. *Healthcare (Basel)* 2025; **14:** 35.

10 Rosenstock J, Allison D, Birkenfeld AL, et al, and the PIONEER 3 Investigators. Effect of additional oral semaglutide *vs* sitagliptin on glycated hemoglobin in adults with type 2 diabetes uncontrolled with metformin alone or with sulfonylurea: the PIONEER 3 randomized clinical trial. *JAMA* 2019; **321:** 1466–80.

11 McGuire DK, Marx N, Mulvagh SL, et al, and the SOUL Study Group. Oral semaglutide and cardiovascular outcomes in high-risk type 2 diabetes. *N Engl J Med* 2025; **392:** 2001–12.

12 Rosenstock J, Hsia S, Nevarez Ruiz L, et al, and the ACHIEVE-1 Trial Investigators. Orforglipron, an oral small-molecule GLP-1 receptor agonist, in early type 2 diabetes. *N Engl J Med* 2025; **393:** 1065–76.

13 Horn DB, Ryan DH, Kis SG, et al, and the ATTAIN-2 Trial Investigators. Orforglipron, an oral small-molecule GLP-1 receptor agonist, for the treatment of obesity in people with type 2 diabetes (ATTAIN-2): a phase 3, double-blind, randomised, multicentre, placebo-controlled trial. *Lancet* 2025; **406:** 2927–44.

14 Wharton S, Aronne LJ, Stefanski A, et al, and the ATTAIN-1 Trial Investigators. Orforglipron, an oral small-molecule GLP-1 receptor agonist for obesity treatment. *N Engl J Med* 2025; **393:** 1796–806.

15 Buse JB, Bajaj HS, Dalskov S-M, et al, and the REIMAGINE 2 study group. Cagrilintide–semaglutide (CagriSema) versus semaglutide or cagrilintide in people with type 2 diabetes (REIMAGINE 2): a double-blind, randomised, controlled, phase 3 study. *Lancet Diabetes Endocrinol* 2026; **14:** 662–77.

16 Lau DCW, Erichsen L, Francisco AM, et al. Once-weekly cagrilintide for weight management in people with overweight and obesity: a multicentre, randomised, double-blind, placebo-controlled and active-controlled, dose-finding phase 2 trial. *Lancet* 2021; **398:** 2160–72.

17 Hollander PA, Levy P, Fineman MS, et al. Pramlintide as an adjunct to insulin therapy improves long-term glycemic and weight control in patients with type 2 diabetes: a 1-year randomized controlled trial. *Diabetes Care* 2003; **26:** 784–90.

18 Dahl K, Toubro S, Dey S, et al. Amycretin, a novel, unimolecular GLP-1 and amylin receptor agonist administered subcutaneously: results from a phase 1b/2a randomised controlled study. *Lancet* 2025; **406:** 149–62.

19 Gasiorek A, Heydorn A, Gabery S, et al. Safety, tolerability, pharmacokinetics, and pharmacodynamics of the first-in-class GLP-1 and amylin receptor agonist, amycretin: a first-in-human, phase 1, double-blind, randomised, placebo-controlled trial. *Lancet* 2025; **406:** 135–48.

20 International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. Guideline for Good Clinical Practice E6 (R3). 2025. https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf (accessed March 30, 2026).

21 World Medical Association. WMA Declaration of Helsinki—ethical principles for medical research involving human participants. 2024. https://www.wma.net/policies-post/wma-declaration-of-helsinki/ (accessed March 30, 2026).

22 Mora P, Aroda VR, Asong M, et al. Efficacy and safety of once-weekly subcutaneous zenagamtide, a novel unimolecular GLP-1 and amylin receptor agonist, in type 2 diabetes: a multicentre, randomised, parallel, double-blind, placebo-controlled, dose-finding, phase 2 trial. *Lancet* 2026; published online July 30. https://doi.org/10.1016/S0140-6736(26)01248-1.

23 Inker LA, Eneanya ND, Coresh J, et al, and the Chronic Kidney Disease Epidemiology Collaboration. New creatinine- and cystatin C–based equations to estimate GFR without race. *N Engl J Med* 2021; **385:** 1737–49.

24 Frias JP, Deenadayalan S, Erichsen L, et al. Efficacy and safety of co-administered once-weekly cagrilintide 2·4 mg with once-weekly semaglutide 2·4 mg in type 2 diabetes: a multicentre, randomised, double-blind, active-controlled, phase 2 trial. *Lancet* 2023; **402:** 720–30.

24 Davies M, Pieber TR, Hartoft-Nielsen M-L, Hansen OKH, Jabbour S, Rosenstock J. Effect of oral semaglutide compared with placebo and subcutaneous semaglutide on glycemic control in patients with type 2 diabetes: a randomized clinical trial. *JAMA* 2017; **318:** 1460–70.

26 Rosenstock J, Yabe D, Cox D, et al, and the ACHIEVE-3 Investigators. Efficacy and safety of once-daily oral orforglipron compared with oral semaglutide in adults with type 2 diabetes (ACHIEVE-3): a multinational, multicentre, non-inferiority, open-label, randomised, phase 3 trial. *Lancet* 2026; **407:** 1147–60.

27 Battelino T, Alexander CM, Amiel SA, et al. Continuous glucose monitoring and metrics for clinical trials: an international consensus statement. *Lancet Diabetes Endocrinol* 2023; **11:** 42–57.

28 Ryan DH, Yockey SR. Weight loss and improvement in comorbidity: differences at 5%, 10%, 15%, and over. *Curr Obes Rep* 2017; **6:** 187–94.

29 Lingvay I, Sumithran P, Cohen RV, le Roux CW. Obesity management as a primary treatment goal for type 2 diabetes: time to reframe the conversation. *Lancet* 2022; **399:** 394–405.

30 Eli Lilly and Company. Lilly's oral GLP-1, orforglipron, superior to oral semaglutide in head-to-head trial. 2025. https://investor.lilly.com/news-releases/news-release-details/lillys-oral-glp-1-orforglipron-superior-oral-semaglutide-head (accessed Feb 11, 2026).

31 Wharton S, Lingvay I, Bogdanski P, et al, and the OASIS 4 Study Group. Oral semaglutide at a dose of 25 mg in adults with overweight or obesity. *N Engl J Med* 2025; **393:** 1077–87.

32 Bays HE. Why does type 2 diabetes mellitus impair weight reduction in patients with obesity? A review. *Obes Pillars* 2023; **7:** 100076.

33 Alexander GC, Xiao X, Dilek S, et al. Heterogeneity of treatment effects of glucagon-like peptide-1 receptor agonists for weight loss in adults: a systematic review and meta-analysis. *JAMA Intern Med* 2026; **186:** 567–77.

34 Frías JP, Auerbach P, Bajaj HS, et al. Efficacy and safety of once-weekly semaglutide 2·0 mg versus 1·0 mg in patients with type 2 diabetes (SUSTAIN FORTE): a double-blind, randomised, phase 3B trial. *Lancet Diabetes Endocrinol* 2021; **9:** 563–74.

35 Frías JP, Davies MJ, Rosenstock J, et al, and the SURPASS-2 Investigators. Tirzepatide versus semaglutide once weekly in patients with type 2 diabetes. *N Engl J Med* 2021; **385:** 503–15.